DHBW Ravensburg

Intelligence Engineering

Solving Real World Problems At Scale

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LECTURE _

Orientation

what this course is, what it assumes, what yor are going to know after.

You have been given a lot of raw **data**,

and you want to engineer the shit out of it.

You have been given a business problem,

and you want to choose the right metrics and objectives to solve it.

Your initial models perform well in offline experiments,

and you want to deploy them at scale.

You want to harden your deployments

to quickly detect, debug, and address issues that arise in production.

There is a lot of manual work and handling involved so far,

and you want to operation vacation your machine learning product or service.

You want to learn about a framework and foundation for building machine learning systems

that can be shared and reused across machine learning use cases and data products.

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Overview

Machine Learning and Data Engineering

The lecture in one slide

Lecture	Contents
I	Overview of Machine Learning Systems
II	Introduction to Machine Learning Systems Design
III	Data Engineering Fundamentals
IV	Training Data
V	Feature Engineering
VI	Model Development and Offline Evaluation
VII	Model Deployment and Prediction Service
VIII	Data Distribution Shifts and Monitoring
IX	Continual Learning and Test in Production
X	Infrastructure and Tooling for MLOps
XI	The Human Side of Machine Learning

The lecture in one slide

I Overview of Machine Learning Systems

- The small part and the big picture
- Understanding Machine Learning Systems

II Introduction to Machine Learning Systems Design

- Business and ML Objectives
- Requirements for ML Systems
- Iterative Process
- Framing ML Problems
- Mind Versus Data

III Data Engineering Fundamentals

- Data Sources
- Data Formats
- Data Models
- Data Storage Engines and Processing
- Modes of Dataflow
- Batch Processing Versus Stream Processing

The lecture in one slide

IV Training Data

Sampling
Labeling
Class Imbalance
Data Augmentation

V Feature Engineering

Learned Features Versus Engineered Features
Common Feature Engineering Operations
Data Leakage
Engineering Good Features

VI Model Development and Offline Evaluation

Model Development and Training
Model Offline Evaluation

The lecture in one slide

VII Model Deployment and Prediction Service

Machine Learning Deployment Myths

Batch Prediction Versus Online Prediction

Model Compression

ML on the Cloud and on the Edge

VIII Data Distribution Shifts and Monitoring

Causes of ML System Failures

Data Distribution Shifts

Monitoring and Observability

IX Continual Learning and Test in Production

Continual Learning

Test in Production

The lecture in one slide

X Infrastructure and Tooling for MLOps

- Storage and Compute
- Development Environment
- Resource Management
- ML Platform
- Build Versus Buy

XI The Human Side of Machine Learning

- User Experience
- Team Structure
- Responsible AI

Course Work

Presentation (15 min) + Q&A (open)

You get recordings of a bee hive entrance, and you are to contribute by data engineering, an annotation policy, a prediction service or specialized tooling.

Open whiteboard, three slides or one page only. Be ready when the class starts.



The bee

Course Work

What I am looking for ..

Introduction by Motivation and Literature

Method by Explanation and or Demonstration

Results show what was achieved and how well

Discussion on findings, limitations and what's next

Smart and Aesthetic Thoughts, Designs, Ideas and Questions.



The bee

Extra Mile

Special projects

If you are in for an extra mile, let me know. I have special projects that I can assign to you.

Wrapped in the same presentation and Q&A format, yes it will be about Bananas, and yes it will be more challenging, more time and effort.

A classic: lose-lose in the short term, yet **win-win in the long term**.



A large banana

Brainstorm together

Brainwriting in the big group, 90 minutes

1. **10 min** Research: look it up online or ask a colleague.
2. **20 min** Each player writes four ideas.
3. **5 min** Research.
4. **10 min** Stacks pass to the next table, each adds two.
5. **5 min** Research.
6. **10 min** Stacks pass again, each adds the last two.
7. **15 min** Cluster and clean the ideas per group.
8. **15 min** Cluster across teams.

Eight ideas per player, every stack read at three tables.



The bee



It is your obligation to

get yourself a presentation slot and brainstorm a project idea to work on.

References

1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.